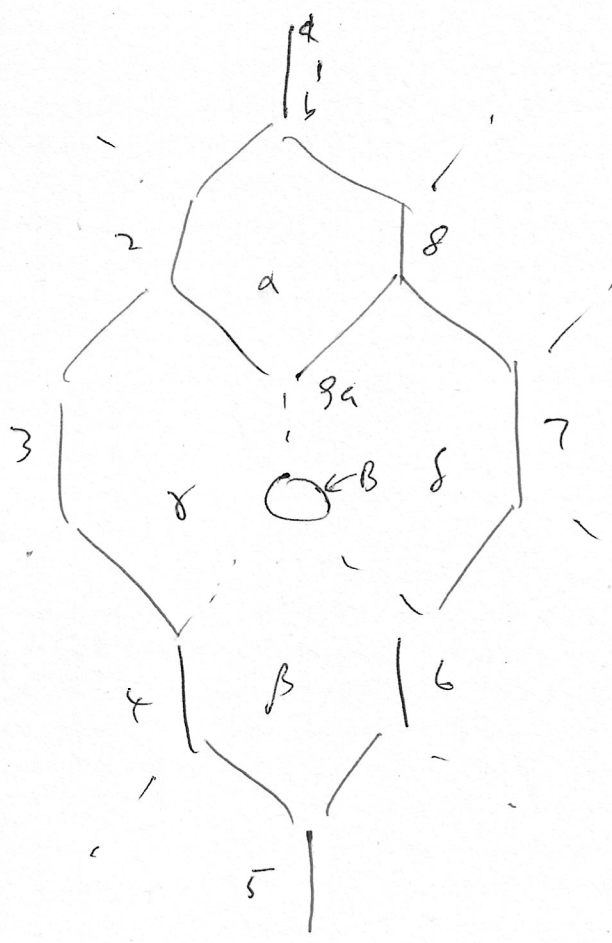


Case with odd numbers inside

Example modified § MS8.2



$$B = ga$$

A has 17 spins

AB has 18 spins

$$\hat{P}_{AB} = \frac{1}{2^6} \left(\frac{1+w_a}{2} \right) \left(\frac{1+w_b}{2} \right) \left(\frac{1+w_c}{2} \right) \left(\frac{1+w_d}{2} \right)$$

$$\left\{ \frac{1}{2^8} (1 + \langle \dim_{\mathbb{R}} A \rangle (0,0) + \dots \right\}$$

... (0,0)

$$W = W_\alpha W_\beta W_\gamma W_\delta \quad \text{entirely in } A$$

$$W_\alpha \quad \text{entirely in } A.$$

$$W_\beta = W'_\beta \sigma_B^z$$

$$W_\gamma = W'_\gamma \sigma_B^y$$

$$W_\delta = W'_\delta \sigma_B^x$$

$$W'_\beta, W'_\gamma, W'_\delta \text{ entirely in } A.$$

(10.2)

rewrite (10.1) as

$$\hat{\rho}_{AB} = \frac{1}{2^4} \left(\frac{1+W}{2} \right) \left(\frac{1+W_\alpha}{2} \right) \left[\frac{1 + W'_\beta \sigma_B^z + W'_\gamma \sigma_B^y + W'_\delta \sigma_B^x}{2} \right]$$

$$\left\{ \frac{1}{2^3} (1 + \langle \text{dim}(\hat{\rho}_{AB}) \rangle \sigma \sigma) + \dots \right\}$$

(10.3)

check: if expanded, single powers $W_\alpha, W_\beta, W_\gamma, W_\delta$ included once

quadratic terms: explicitly $W_\alpha W_\beta, W_\alpha W_\gamma, W_\alpha W_\delta$

$$\text{also } W(W_\alpha W_\beta) = W_\gamma W_\delta$$

$$W(W_\alpha W_\gamma) = W_\beta W_\delta$$

$$W(W_\alpha W_\delta) = W_\beta W_\gamma$$

cubic $W W_\alpha W_\beta W_\gamma W_\delta$

quartic $= W$

The last factor contains

$$\left\{ \frac{1}{2^8} \left(1 + \langle \sigma_{9a}^z \sigma_B^z \rangle \sigma_{9a}^z \sigma_B^z + \langle \sigma_{6a}^x \sigma_B^x \rangle \sigma_{6a}^x \sigma_B^x \right. \right. \\ \left. \left. + \langle \sigma_{6a}^y \sigma_B^y \rangle \sigma_{6a}^y \sigma_B^y \right. \right. \\ \left. \left. + (\text{dimers insertions involving } A \text{ only}) \right\}$$

$$\hat{\rho}_A = \text{Tr}_B \hat{\rho}_{AB}$$

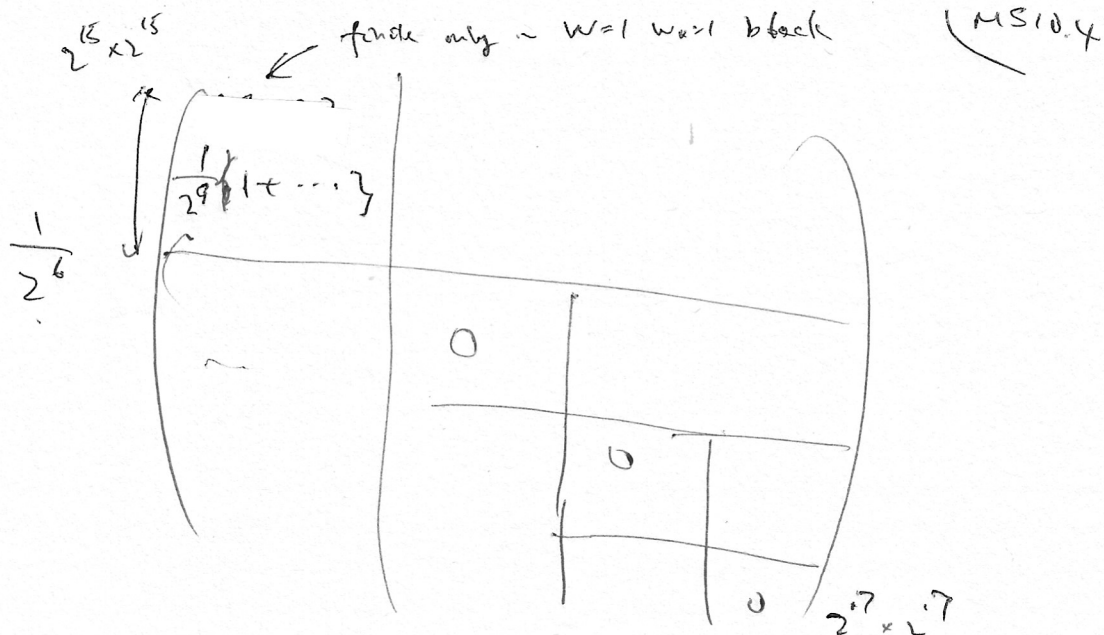
$$= \frac{1}{2^6} \left(\frac{1+W}{2} \right) \left(\frac{1+W_A}{2} \right) \left\{ \frac{1}{2^2} \left(1 + \langle \sigma_{9a}^z \sigma_B^z \rangle W'_\beta \sigma_{9a}^z \right. \right. \\ \left. \left. + \langle \sigma_{6a}^x \sigma_B^x \rangle W'_\delta \sigma_{6a}^x + \langle \sigma_{6a}^y \sigma_B^y \rangle W'_\gamma \sigma_{6a}^y \right. \right. \\ \left. \left. + (\text{dimer insertions}) \right\} \quad (10.4)$$

$$\text{eg } \langle \sigma_{2c}^z \sigma_{2c}^z \sigma_{9c}^z \sigma_B^z \rangle W'_\beta \sigma_{9a}^z \sigma_{2c}^z \sigma_{2c}^z \dots$$

is purely in A

generally $\langle \mathcal{D} \rangle_D$ $\langle \sigma_{9a}^z \sigma_B^z \mathcal{D} \rangle (W'_\beta \sigma_{9a}^z \mathcal{D})$ etc.
with $\mathcal{D}$ in A.

structure:

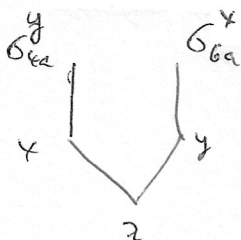


... (10.5)

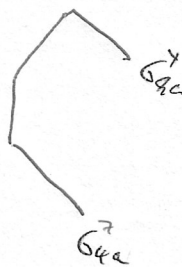
cf. (8.10)

note

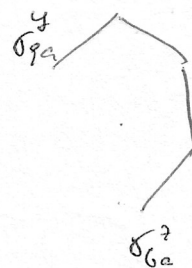
$$W'_\beta =$$



$$W'_\gamma =$$



$$W'_\delta =$$



anticommute with each other

Special case,

2-dimer limit, then

$$\uparrow \} \rightarrow \left\{ \frac{1}{2^8} \left(1 + W'_\beta \sigma_{9a}^z + \left(\begin{array}{c} \text{dimer incentive} \\ \sigma^+ \sigma^+ \text{ only} \end{array} \right) \right) \right.$$

$$= \frac{1}{2} \left(1 + W'_\beta \sigma_{9a}^z \underbrace{\left(\frac{1 + (\sigma^+ \sigma^+)}{2} \right)}_{\text{8 pairs}} \right)$$

= projectors only

If diagonalized, then

structure: $\frac{1}{2^6} \begin{pmatrix} 1 & & & \\ & 0 & & \\ & & \ddots & \\ & & & 0 \end{pmatrix}$

$$S = 6 \ln 2$$

$$\left[L_{\text{out}} = 6 \right]$$

if cut out also $9a$, $L_{\text{in}} = 2$

$$S = ((6-1) + (2-1)) \ln 2 = 6 \ln 2$$

got same entropy

cutting out further 2e isolated $9a$

does not change EE, as expected

Generally,

$$\{ \} = \left\{ \frac{1}{2g} (1 + \dots) \right\}$$

(eigenvalues $\neq 0, \neq 1$)

is no longer a projector and contributes

extra term to S